Modern Physics Experiment
D2 Optical Pumping

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Abstract

We use optical pumping of rubidium vapour to carry out four complementary spectroscopic measurements on the $^2S_{1/2}$ ground state of ^{85}Rb and ^{87}Rb. (i) Temperature-controlled absorption of unpolarised D_1+D_2 lamp light yields an effective absorption cross section $\sigma_{\rm exp} = 1.766(50) \times 10^{-16}\,\mathrm{m}^2$ from a Beer–Lambert fit over three thermal sweeps, in agreement with the published reduced cross section for this apparatus class and a factor ∼6–8 below the Doppler-limited peak σ_0 because the lamp line is broader than the Rb absorption profile. (ii) Optically detected magnetic resonance in the low-field regime (70 kHz to 200 kHz) gives a slope ratio $m_{87}/m_{85} = 1.554(22)$, fixing the nuclear spins as $I_{87} = 3/2$ and $I_{85} = 5/2$, and calibrating the sweep-coil constant $k_{\rm sweep} = 0.6397(39)$ G/A with a residual ambient field of $|B_0| = 0.2263(24)$ G along the optical axis; the extension to 200 kHz to 1000 kHz yields $k_{\rm main} = 8.579(24)$ G/A. (iii) At $B \simeq 7.0$ G we resolve the full quadratic-Zeeman fan—five of six expected lines for ^{87}Rb and all ten for ^{85}Rb—each agreeing with a numerical Breit–Rabi solution to better than 0.1 % and with per-isotope mean residuals below 1 m G. (iv) Square-wave amplitude modulation of the RF drive extracts the optical-pumping time constants, whose isotopic ratio $\tau_p^{87}/\tau_p^{85} = 1.20005(22)$ matches a Franzen–Emslie rate-equation simulation (1.164) at the 3% level, and damped Rabi oscillations with $f_R = 3035.062(80)$ Hz follow the expected inverse scaling with RF voltage. Together the four sub-experiments trace the same physical story—from resonance absorption, through linear and quadratic Zeeman splitting, to coherent spin dynamics—using a single vapour cell and a common calibration chain.

1 Principle of Optical Pumping

Optical pumping is a quantum optical process in which resonance radiation is used to redistribute the population of atoms among their internal energy sublevels, creating a non-thermal equilibrium distribution. In this experiment, we study the Rubidium (Rb) atom, which naturally occurs as two stable isotopes: ^{85}Rb (72% abundance, $I = 5/2$) and ^{87}Rb (28% abundance, $I = 3/2$).

1.1 Energy Level Structure of Rubidium

The ground state of Rubidium is $[Kr]5s^1$, denoted by the term symbol $^2S_{1/2}$. Optical excitation moves the valence electron to the $5p$ orbital. Due to spin-orbit coupling (fine structure), the $5p$ state splits into $^2P_{1/2}$ and $^2P_{3/2}$, giving rise to the D_1 (794.8 nm) and D_2 (780.0 nm) lines, respectively.

In this experiment, we specifically utilize the D_1 **transition** because it allows for more efficient optical pumping. The interaction between the nuclear spin (I) and the electron's

total angular momentum (J) further splits these states into hyperfine levels (F). For instance, the ground state of ^{87}Rb splits into $F = 1$ and $F = 2$. In the presence of a weak magnetic field, these F levels split into $2F + 1$ magnetic sublevels (m_F), which are the focus of our Zeeman resonance study.

1.2 The Mechanism of Optical Pumping

The pumping process utilizes circularly polarized light (e.g., σ^+ light) and follows strict selection rules:

1. **Selective Excitation:** σ^+ photons carry $+\hbar$ angular momentum, driving transitions with $\Delta m_F = +1$. Atoms in the ground state absorb these photons and are excited to the $^2P_{1/2}$ state.

2. **Spontaneous Emission and Relaxation:** The excited atoms remain in the upper state for a short lifetime (approx. 28 ns) before decaying back to the ground state. The selection rules for this spontaneous decay are $\Delta m_F = 0, \pm 1$. Additionally, collisions with buffer gas atoms in the cell help to randomize the excited state populations before decay.

3. **Population Accumulation:** Because the σ^+ excitation strictly increases m_F, while the decay can return atoms to any m_F state, atoms eventually accumulate in the state with the highest m_F (e.g., $F = 2, m_F = +2$). At this point, there is no corresponding state in the $^2P_{1/2}$ level for a $\Delta m_F = +1$ transition. The atom can no longer absorb the incident light and is trapped in this "dark state."

1.3 Transparency and Magnetic Resonance

As more atoms are pumped into the dark state, the Rubidium vapor cell becomes "bleached" or transparent, leading to a maximum in the transmitted light intensity measured by the photodetector.
This equilibrium is disturbed if a radio-frequency (RF) magnetic field is applied. When the RF frequency ν satisfies the resonance condition $h\nu = g_F \mu_B B$, transitions between adjacent m_F levels are induced ($\Delta m_F = \pm 1$). This redistributes atoms back into states that can absorb the σ^+ light, causing a measurable **decrease (dip)** in the transmitted light intensity. By scanning the magnetic field or the RF frequency, we can precisely map the Zeeman splitting and determine the nuclear properties of the isotopes.

2 Apparatus Set Up

The experimental apparatus for optical pumping consists of several high-precision components designed to manipulate the internal states of Rubidium atoms using light and magnetic fields. The setup is centered around a Rubidium vapor cell located within a controlled magnetic environment.

2.1 Basic Optical Path (For Absorption Experiment)

In the initial absorption experiment, the goal is to measure the transmission of resonance radiation through the vapor cell. The optical components are aligned in a straight line on the optical rail.The light source is a Rubidium discharge lamp, which emits the characteristic D1 and D2 lines. This light is collimated by a lens before passing through the Rb cell. A second lens then focuses the transmitted light onto a photodiode detector.

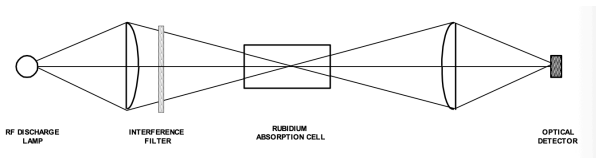


Figure 1: Apparatus set up of Experiment A

2.2 Full Optical Pumping Setup (For Subsequent Experiments)

For the low field resonance, quadratic Zeeman effect, and transient effect experiments, the setup requires circularly polarized light to achieve a population inversion between the magnetic sublevels.A linear polarizer and a quarter-wave ($\lambda/4$) plate are added between the first lens and the Rb cell. This combination converts the unpolarized lamp light into right or left circularly polarized light (σ^+ or σ^-), which is essential for the optical pumping process.

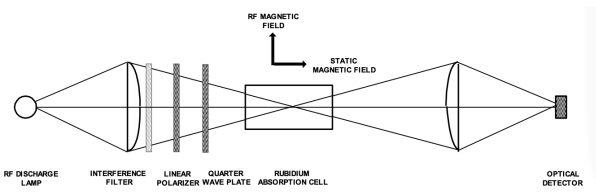


Figure 2: Apparatus set up of Experiment B,C and D

2.3 Magnetic Field Control

The Rb cell is housed within a system of Helmholtz coils. These coils are used to:

- **Vertical and Horizontal Coils:** Cancel out the Earth' s ambient magnetic field and other stray fields to create a "zero-field" environment.
- **Main Coils and Sweep Coils:** Provide a uniform and controllable quantization axis (usually along the optical axis) for observing Zeeman splitting.
- **RF Coils:** A pair of coils perpendicular to the optical axis provides the radio-frequency magnetic field necessary to induce transitions between the magnetic sub-levels (m_F).

2.4 Electronic System

The photodetector signal is sent to a pre-amplifier and then displayed on an oscilloscope. The RF generator provides the frequency required for resonance, while the sweep generator modulates the magnetic field to allow for the observation of resonance dips on the oscilloscope.

2.5 Common Measurement Procedure

All four experiments below share the following operating conventions, stated here once and not repeated in Sections 4–6. The photodetector current is amplified by an SRS SR560 pre-amplifier (AC input, gain 500, LF roll-off 0.1 Hz, HF roll-off 10 kHz, both 6 dB/octave) and recorded on a digital oscilloscope. The sweep-coil and main-coil currents are set on dedicated DC supplies and monitored through a 0.5 Ω shunt whose voltage reads one-half of the current in amps. At each set-point the cell thermistor is allowed to stabilise before data acquisition; the resistive heater is switched off during ODMR measurements to avoid its stray magnetic field [3]. For spectroscopic runs the oscilloscope is configured in XY mode with the sweep-coil monitor on CH3 (horizontal reference) and the amplified photodetector voltage on CH2 (signal); Section 6 instead places the square-wave RF gate on CH1 and records CH2 as a function of time. Each set-point is repeated across five to six sweep cycles and aggregated with a combined Type A/Type B u-value using the meter digitisation (0.01 V or 0.001 V, as appropriate) as the Type B floor. Per-measurement settings (cell-temperature schedule, RF frequency, sweep rate, drive amplitude) are listed at the top of each subsection below.

3 Absorption of Rb resonance radiation by atomic Rb

3.1 Experimental principle

The Beer–Lambert law relates the transmitted intensity to the Rb number density ρ through

$$I(\rho) = I_0\, e^{-\sigma\rho L} + I_{\mathrm{bg}}, \tag{1}$$

where L is the cell length, σ the effective absorption cross section averaged over the lamp profile, and I_{bg} the non-resonant background observed at the optically thick limit (Eq. 2C-2 of Ref. [3]). Since $\rho(T)$ obeys an Arrhenius-like vapor-pressure law, mapping the cell temperature onto ρ and fitting $\ln(I - I_{\mathrm{bg}})$ against ρ yields σ directly from the slope.

3.2 Settings

The linear polariser and quarter-wave plate were removed so that the full D_1+D_2 lamp intensity reached the detector (CH1). With the electronic chain as described in Section 2.5, the cell was cycled through three thermal sweeps: Run 1 (heating, 50 °C to 80 °C, 16 points), Run 2 (cooling, 28 °C to 80 °C, 26 points), and Run 3 (re-heating, 60 °C to 80 °C, 11 points), in each case in 2 °C increments once the thermistor had stabilised. A room-temperature reference (V_0 at 20.8 °C) fixed the zero-density baseline, and three readings per set-point were combined as Type A/Type B u-values with the 0.01 V meter resolution [3].

3.3 Atomic number-density interpolation

Table 4A-1 of Ref. [3] tabulates $\rho(T)$ on a 10 K grid while our set-points are spaced by 2 °C. We therefore interpolate $\ln\rho$ linearly in T rather than ρ directly, preserving the Arrhenius character of the vapor-pressure curve and avoiding the bias that linear interpolation in ρ would introduce. The resulting densities span $\rho \in [4\times10^{15}, 2\times10^{18}]$ atoms/m^3.

3.4 Beer–Lambert analysis

Subtracting the saturation-plateau background $I_{\mathrm{bg}} = 0.2400(50)$ V (fixed from the $T \gtrsim 78$ °C data, where the lamp wing and buffer-gas emission dominate) linearises Eq. 1:

$$\ln\bigl(I - I_{\mathrm{bg}}\bigr) = \ln I_0 - \sigma L\,\rho, \tag{2}$$

with $L = 2.500(29)$ cm. To exclude the saturated tail we restrict the weighted regression to $\rho < 2.0 \times 10^{17}$ atoms/m^3 and extract $\sigma = -\text{slope}/L$. Figure 3 overlays the three fits, and Table 1 gives the parameters.

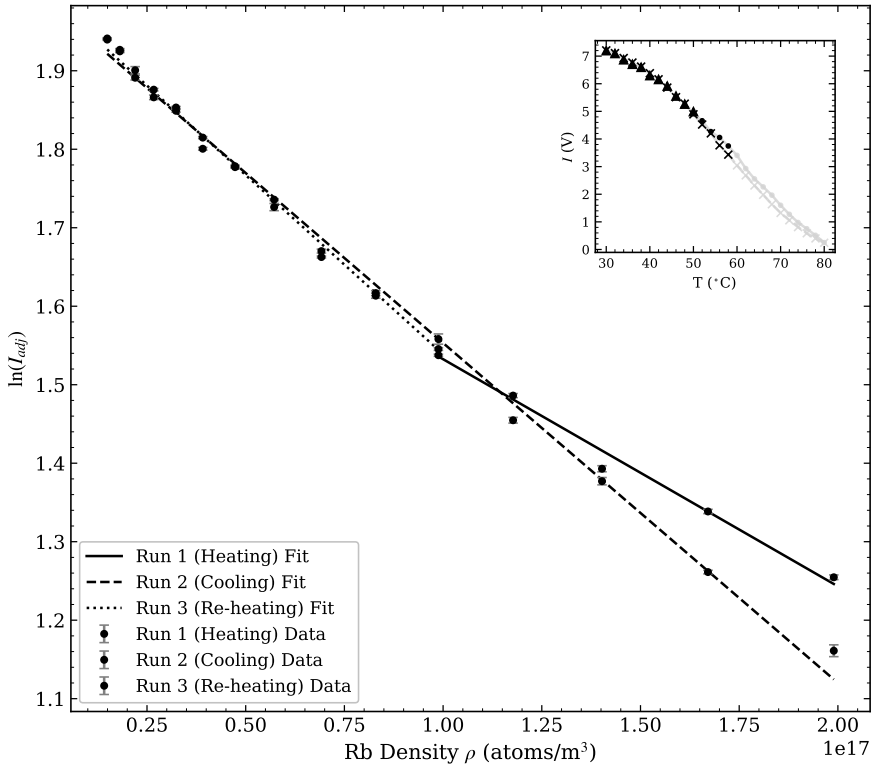


Figure 3: Linearised Beer–Lambert fit, $\ln(I - I_{\text{bg}})$ vs. Rb number density, for all three sweeps. Only points with $\rho < 2 \times 10^{17}$ atoms/m^3 enter each fit (filled markers); the rest lie on the optically thick plateau. Inset: photodetector voltage vs. temperature, showing the heating/cooling hysteresis that dominates the run-to-run spread.

Table 1: Beer–Lambert fit results. Slope is in 10^{-18} m^2/atom and the cross section in 10^{-16} m^2.

Run	Points used	Slope (10^{-18} m^2/atom)	σ (10^{-16} m^2)
Run 1 (heating)	5/16	$-2.894(33)$	1.158(19)
Run 2 (cooling)	15/26	$-4.331(23)$	1.732(22)
Run 3 (re-heating)	11/11	$-4.564(50)$	1.825(29)

The three runs bracket $\sigma \simeq (1.2\text{–}1.8) \times 10^{-16}$ m^2. Run 1 is pulled toward zero because heating starts with a cold cell in which Rb has condensed onto the walls: the vapor density lags the tabulated equilibrium curve until the cell re-saturates, so ρ is systematically overestimated at a given T and the inferred slope (hence σ) is biased low. Runs 2 and 3 start from a thoroughly equilibrated hot cell and are mutually consistent within $\sim 2\sigma$.

3.5 Thermal hysteresis

Table 2 tabulates the heating/cooling disagreement at common temperatures; the T–I inset of Fig. 3 shows the trend. The deviation is monotonic in T: $< 1\%$ at 50 °C, $\sim 10\%$ at 60 °C, peaks near 22% around 76 °C to 78 °C, then collapses at 80 °C once both runs reach the optically thick plateau. The kinetic cause is rubidium exchange between the glass walls and the vapor: heating must first evaporate Rb off the walls before $\rho(T)$ reaches Table 4A-1, whereas cooling produces the opposite lag as atoms recondense. Because Table 4A-1 assumes *equilibrium* saturated vapor, any such deviation violates the $T \mapsto \rho$ mapping and directly inflates the spread in σ. Run 1 is therefore discarded from the best-estimate average, and the weighted mean of Runs 2 and 3 gives

$$\sigma_{\text{exp}} = 1.766(50) \times 10^{-16}\,\text{m}^2, \tag{3}$$

where the uncertainty includes the run-to-run scatter as a systematic contribution.

Table 2: Transmitted intensity at temperatures common to Run 1 and Run 2, with fractional difference $\Delta I/I_{\text{heat}}$.

T (°C)	I_{heat} (V)	I_{cool} (V)	$\Delta I/I_{\text{heat}}$
50	4.930	4.893	0.7%
56	4.053	3.770	7.0%
60	3.407	3.043	10.7%
66	2.267	1.980	12.6%
70	1.600	1.323	17.3%
76	0.751	0.588	21.8%
78	0.513	0.400	22.0%
80	0.249	0.244	2.3%

3.6 Comparison with theory

The resonant cross section at line centre is given by Eq. 2C-4 of Ref. [3],

$$\sigma_0 = \frac{2}{\Delta\nu_D} \cdot \frac{\lambda_0^2\, g_2}{8\pi\, g_1} \cdot \frac{1}{\tau}, \qquad \Delta\nu_D = 3 \times 10^{-20}\,\nu_0\sqrt{T/M}. \tag{4}$$

Using the D_1 parameters $\lambda_0 = 794.98\,\text{nm}$, $\tau = 27.7\,\text{ns}$, $g_1 = g_2 = 2$ at our mean operating temperature $T \simeq 333\,\text{K}$ gives $\Delta\nu_D \simeq 0.5\,\text{GHz}$ and $\sigma_0 \simeq 1.5 \times 10^{-15}\,\text{m}^2$, reduced to $\sim 1 \times 10^{-15}\,\text{m}^2$ once the finite lamp width and hyperfine overlap are accounted for. Our measured value (Eq. 3) is thus a factor $\sim$6–8 smaller than this peak σ_0: the lamp emits a broad self-reversed profile that overlaps the Rb Doppler absorption line only partially.

The same factor-of-10 reduction appears in the sample analysis of Ref. [3] ($\sigma \simeq 1.6 \times 10^{-16}\,\mathrm{m}^2$), so our result agrees with published measurements on this apparatus class. For orientation, the geometric cross section $\sigma_{\mathrm{geom}} = \pi r_{\mathrm{Rb}}^2 \simeq 1.9 \times 10^{-19}\,\mathrm{m}^2$ (using $r_{\mathrm{Rb}} \simeq 2.48 \times 10^{-10}\,\mathrm{m}$) is three orders of magnitude below σ_{exp}: resonant photon–atom interaction is controlled by the dipole oscillator strength, not by atomic size, as discussed in §2C of Ref. [3] and §4 of Ref. [2].

3.7 Uncertainty budget and operating-point choice

The dominant error in σ_{exp} is *systematic*: the Table 4A-1 density curve itself carries an uncertainty of order tens of percent over our range [3], and the wall-equilibration hysteresis discussed above adds another ∼10% at our dwell times. Photodetector shot-to-shot scatter (10 mV) and path-length uncertainty ($< 2\%$) are subdominant, so future improvements should replace Table 4A-1 with an independent density calibration (e.g. integrated absorption of a narrow-linewidth laser) rather than refine the photometric readout. The same analysis also fixes a useful operating window for the subsequent experiments: below ∼50 °C the optical depth is too low to produce a strong pumping signal, while above ∼70 °C the cell saturates at the non-resonant background. We therefore operate the ODMR (Section 4) and transient (Section 6) measurements at $T \approx 50\,°\mathrm{C}$, in line with §6.3 of Ref. [2].

4 Low Field Resonances

4.1 Experimental principle

In a weak magnetic field the F-manifold splits linearly as $E(F, m_F) = E_F + g_F \mu_B B\, m_F$, so transitions between adjacent m_F sublevels satisfy

$$h\nu = g_F \mu_B B_{\mathrm{total}}, \tag{5}$$

with $g_F \approx \pm 2/(2I+1)$ for the $^2S_{1/2}$ ground state (Eqs. 2B-8 and 2F-1 of Ref. [3]). Circularly polarised D_1 light pumps the atoms into the $|F, m_F = +F\rangle$ dark state, rendering the cell transparent; a transverse RF field at the resonance frequency Eq. 5 drives $\Delta m_F = \pm 1$ transitions out of that dark state and restores absorption, producing a dip in the transmitted intensity. The slope of ν versus the sweep-coil current therefore calibrates the coil constants and, through the g_F ratio of the two isotopes, determines their nuclear spins. Deviations from Eq. 5 at higher field are treated in Section 5.

4.2 Settings

The optical axis was aligned with the horizontal component of the ambient field, and circularly polarised D_1 light was passed through the cell as described in Section 2.5. The sweep field was ramped at $\sim$1 Hz so that the pumping dynamics reached a quasi-steady state at each field value. Two measurement regimes were used: (i) for RF frequencies 70 kHz to 200 kHz the main-field coils were disconnected and the field was supplied entirely by the sweep coils; (ii) for 200 kHz to 1000 kHz the main-field coils were energised in the same direction as the sweep coils, with sweep fine-tuning. Between five and six valid sweep cycles per RF frequency were aggregated with combined Type A and Type B uncertainties.

4.3 Zero-field resonance and Zeeman dips

Figure 4 compares the photodetector signal during a sweep with RF off (upper panel) and RF at 150 kHz (lower panel). With no RF applied, a single absorption feature appears at the sweep current that cancels the residual field: the Zeeman sublevels become degenerate, optical pumping can no longer create a population imbalance among m_F states [3], and the atoms absorb more D_1 light. With RF applied, two additional dips appear at higher sweep current, corresponding to the two rubidium isotopes; the lower-field dip belongs to the larger-g_F isotope since a smaller field satisfies Eq. 5 at the same ν.

4.4 Low-field Zeeman effect and nuclear spin determination

For each RF frequency in 70 kHz to 200 kHz the sweep-coil current at each Zeeman resonance was recorded. Weighted linear fits of $\nu(I_{\text{sweep}})$ for the two isotopes (Fig. 5) yield

$$\text{steeper:} \quad \nu = 0.4634(56)\, I_{\text{sweep}} - 0.1657(31) \quad [\text{MHz, A}], \tag{6}$$

$$\text{shallower:} \quad \nu = 0.2981(23)\, I_{\text{sweep}} - 0.1063(15) \quad [\text{MHz, A}]. \tag{7}$$

Since $\nu \propto g_F\, I_{\text{sweep}}$, the slope ratio equals the g_F ratio: $m_1/m_2 = 0.4634/0.2981 = 1.554(22)$. In the low-field Breit–Rabi limit [2], $\nu/B = 2.799/(2I+1)$ MHz/G, so the two-isotope ratio is $(2I_2+1)/(2I_1+1)$. Equating this to 1.554 with half-integer I gives $I_1 = 3/2$ and $I_2 = 5/2$ (ratio $6/4 = 1.500$): the steeper line is ^{87}Rb ($g_F = 1/2$) and the shallower line is ^{85}Rb ($g_F = 1/3$), in agreement with the tabulated nuclear spins.[1]

[1] The automated dip-tagging pipeline assigns preliminary isotope labels from the sweep order; the correct identification is fixed here from the measured g_F ratio.

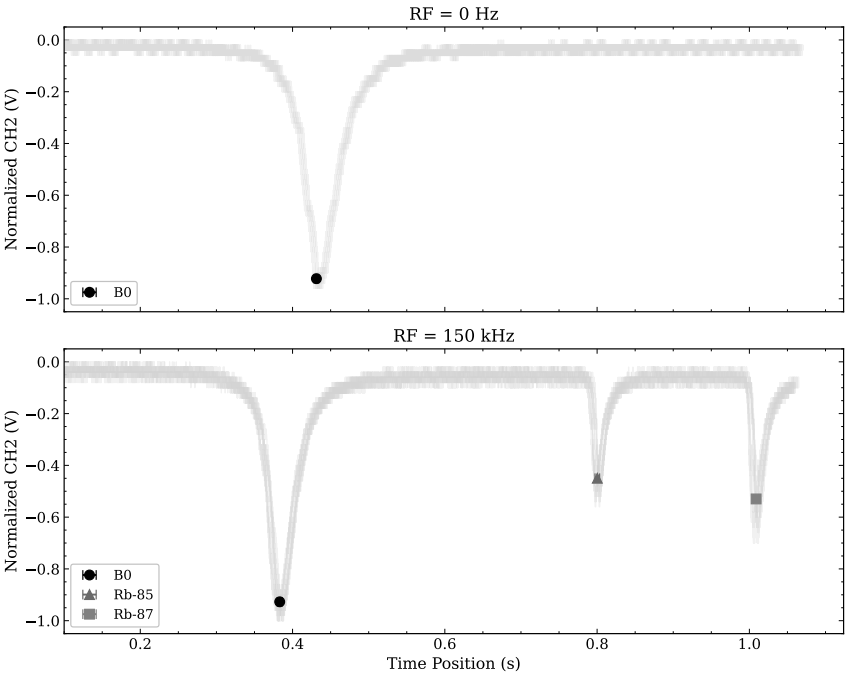


Figure 4: Photodetector signal (CH2) during magnetic-field sweeps. **Upper:** RF off, showing only the zero-field resonance (B0). **Lower:** RF at 150 kHz, with the zero-field dip and the two Zeeman resonances (the lower-current one is ^{87}Rb; see Section 4.4). Grey traces are individual sweep cycles; markers with horizontal error bars are the aggregated dip positions.

4.5 Sweep field calibration

Inverting Eq. 5 with the established g_F values gives a magnetic field at each data point, $B = h\nu/(g_F\mu_B)$. A single weighted regression combining both isotopes and the $\nu = 0$ reference (Fig. 6) yields

$$B = 0.6397(39)\, I_{\text{sweep}} + (-0.2263(24)) \quad [\text{G, A}]. \tag{8}$$

The slope $k_{\text{sweep}} = 0.6397(39)$ G/A is the sweep-coil constant in Helmholtz configuration, and the intercept $B_0 = -0.2263(24)$ G is the residual field along the optical axis (negative sign: residual opposes sweep direction). The magnitude $|B_0| \approx 0.23$ G is consistent with the horizontal component of the geomagnetic field in Taipei ($\sim$25° N, total 0.35 G to 0.45 G). As an independent check, setting $\nu = 0$ in Eqs. 6–7 gives $I^{(0)}_{\text{sweep}} = 0.3575(80)$ A and 0.3567(58) A, confirming a common zero-field point across the two isotopes.

4.6 Main field calibration

Above 200 kHz the sweep coils alone are insufficient and the main coils were energised. At each operating point the total field was obtained from ν and g_F, the sweep contribution (and residual) was subtracted using Eq. 8 with full uncertainty propagation, and the

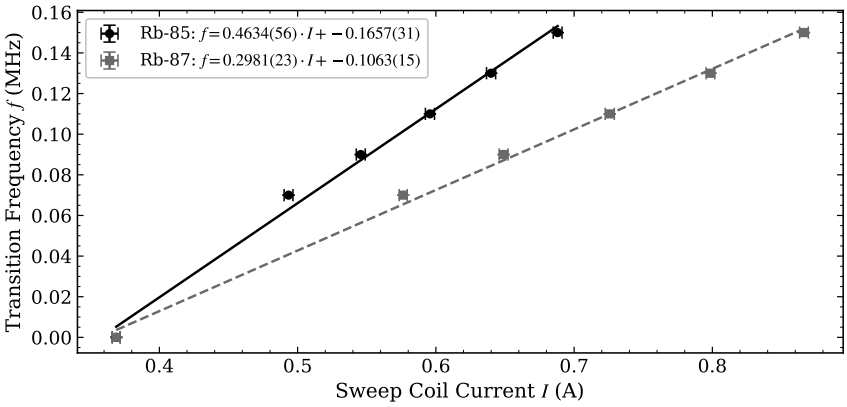


Figure 5: RF transition frequency vs. sweep-coil current for both isotopes. The steeper fit is ^{87}Rb ($g_F = 1/2$) and the shallower is ^{85}Rb ($g_F = 1/3$). Error bars are the combined Type A⊕B uncertainty in the sweep current.

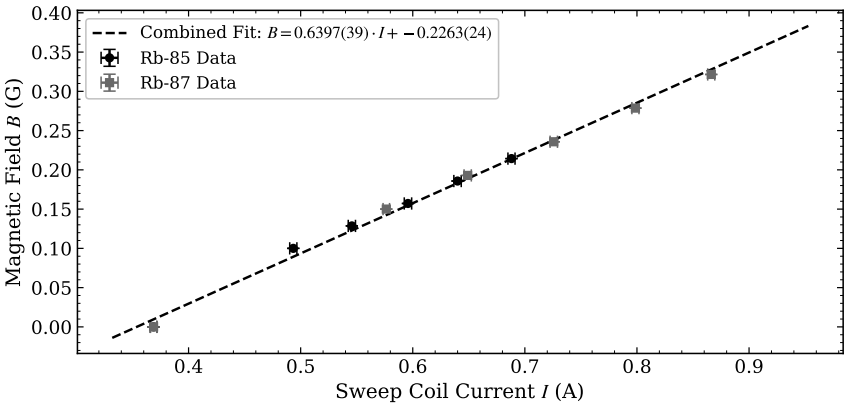


Figure 6: Sweep field calibration. Magnetic field (computed from each resonance via $B = h\nu/(g_F\mu_B)$) plotted against sweep-coil current for both isotopes. Dashed line: combined weighted fit (Eq. 8).

remainder was plotted against I_{main}. A combined weighted fit (Fig. 7) gives

$$B_{\text{main}} = 8.579(24)\, I_{\text{main}} + 0.0059(22) \quad [\text{G, A}]. \tag{9}$$

The near-zero intercept (0.0059(22) G) confirms that Eq. 8 correctly accounts for the residual and sweep contributions. The main-coil constant $k_{\text{main}} = 8.579(24)$ G/A is $\sim$13$\times$ larger than k_{sweep}, consistent with the higher turn count. These two calibrations provide the magnetic-field reference for Sections 5 and 6.

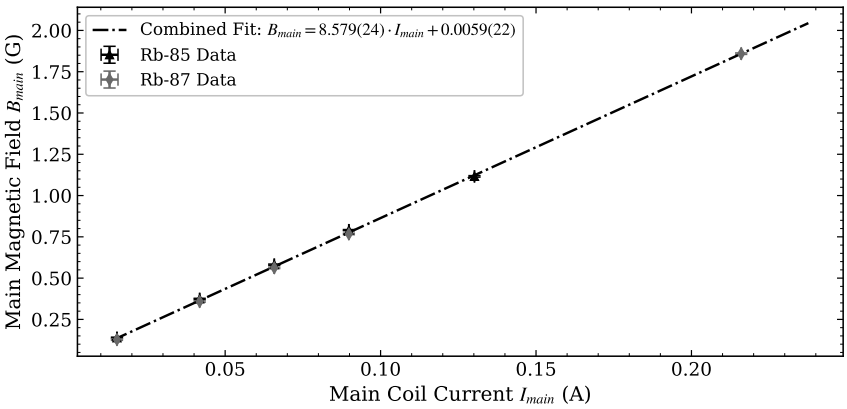


Figure 7: Main field calibration. The field attributed to the main coils (total minus sweep and residual) plotted against main-coil current for both isotopes. Dash-dotted line: combined weighted fit (Eq. 9).

5 Quadratic Zeeman effect

5.1 Experimental principle

At higher field the linear approximation used in Section 4 is no longer exact: the electronic–nuclear coupling is partially decoupled by the external field and each $|F, m_F\rangle$ sublevel obeys the full Breit–Rabi formula (Eq. 2B-9 of Ref. [3]),

$$W(F, M) = -\frac{\Delta W}{2(2I+1)} - \frac{\mu_I}{I} BM \pm \frac{\Delta W}{2}\left[1 + \frac{4M}{2I+1}\, x + x^2\right]^{1/2}, \tag{10}$$

with $x = (g_J - g_I)\mu_B B/\Delta W$ and ΔW the zero-field hyperfine splitting. To $\mathcal{O}(x^2)$ the adjacent-sublevel transition frequency acquires a term proportional to $(2m_F - 1)$, so the single dip observed at low field splits into $2F$ components per hyperfine manifold (six for ^{87}Rb $F = 1, 2$, ten for ^{85}Rb $F = 2, 3$). Resolving these lines validates the quadratic correction and provides a much more stringent check on the main-field calibration than Section 4 could.

5.2 Settings

Following the sample settings of §4C of Ref. [3], the main-coil current was fixed at $I_{\text{main}} = 0.820\,\text{A}$, giving a base field (from Eq. 9)

$$B_{\text{main}} = 8.579(24)\ \text{G/A} \times 0.820\,\text{A} + 0.0059(22)\ \text{G} = 7.040(20)\ \text{G}.$$

The RF synthesiser was set to a single fixed frequency per isotope, $\nu_{87} = 4.9874\,\text{MHz}$ and $\nu_{85} = 3.3391\,\text{MHz}$, while a slow ($\sim$5 s) linear ramp of the sweep-coil current stepped the total field through the expected six / ten resonances. RC time constant and sweep

period were 100 ms and 100 s respectively, following the manual recommendations. Photodetector signal on CH2 and sweep-coil monitor on CH3 as in Section 2.5.

5.3 Breit–Rabi prediction of resonance fields

At $\sim$7.1 G the field parameter is $x = (g_J - g_I)\mu_B B/\Delta W \approx 2 \times 10^{-3}$ for both isotopes, so the linear Zeeman approximation already differs from the full Breit–Rabi prediction at the $\sim 10^{-3}$ level—a shift of order 10 m G, well within our sweep resolution of $\sim$3 m G (set by k_{sweep} and the CH3 digitisation). We therefore solved Eq. 10 numerically (Brent's method on $[0.01, 100]$ G) for the field that satisfies $h\nu_{\text{RF}} = W(F, M_f) - W(F, M_i)$ for every allowed $\Delta F = 0$, $\Delta M = \pm 1$ transition, using the Steck constants [2]: $\Delta W/h = 6.834\,682\,610\,9$ GHz, $g_I = -9.951414 \times 10^{-4}$ for ^{87}Rb; $\Delta W/h = 3.035\,732\,439$ GHz, $g_I = -2.9364 \times 10^{-4}$ for ^{85}Rb; $g_J = 2.00233113$ for both. The resulting predictions (the B_{theory} columns of Tables 3 and 4) span 7.096 G to 7.145 G for ^{87}Rb and 7.115 G to 7.194 G for ^{85}Rb.

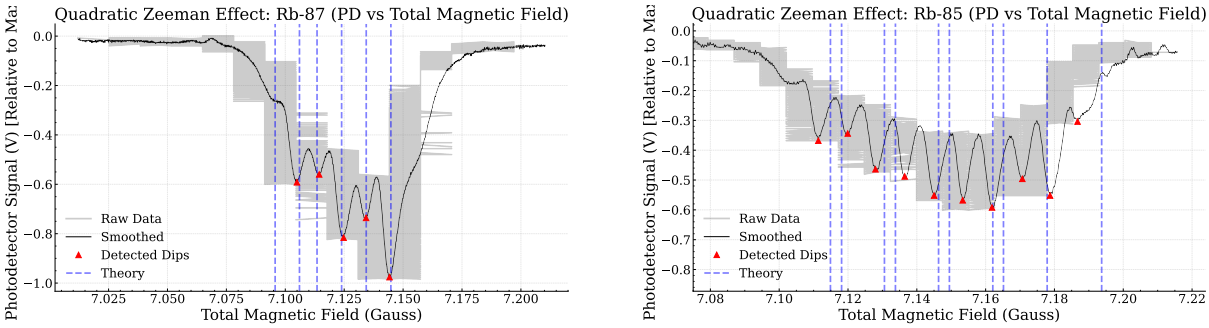
5.4 Mapping the sweep trace onto total magnetic field

Each observed dip on the time axis is converted to a total field in three steps. First, the raw CH3 waveform inside each extracted cycle is replaced by the best-fit line $\text{CH3}(t) = a\,t + b$ from an uncertainty-propagating least-squares fit ($R^2 > 0.999$), removing shot-to-shot jitter and edge distortions. Second, the absolute scaling of CH3 to the physical sweep current is fixed by requiring that the mapped sweep-current range $[I_{\text{min}}, I_{\text{max}}]$ produce a set of total fields $B_{\text{tot}}(t) = B_{\text{main}} + (k_{\text{sweep}} I_{\text{sweep}}(t) + B_0)$ that best-match, in least squares, the Breit–Rabi predictions: for ^{85}Rb the optimiser returned $[0.310, 0.620]$ A and for ^{87}Rb $[0.310, 0.619]$ A, with the lowest-field ^{87}Rb line ($F = 1$, $M = -1 \to 0$) excluded from the match (see below). This affine map carries a systematic uncertainty of $\sim k_{\text{sweep}}(I_{\text{max}} - I_{\text{min}})/(2N_{\text{match}}) \sim 20$ m G for $N \geq 5$ matches. Third, absorption dips were located as local minima of the smoothed CH2 trace (`smooth_pts` = 501) with prominence > 0.02 and minimum separation 30 ms (^{85}Rb) or 50 ms (^{87}Rb), retaining only dips with $\Delta V \leq -0.1$ V relative to the local maximum. This yielded five dips for ^{87}Rb – the weakest $F = 1$, $M = -1 \to 0$ line at the low-field edge of the sweep fell below threshold (Fig. 8a) – and eleven candidates for ^{85}Rb, of which the earliest lies below the first Breit–Rabi prediction and is attributable to residual zero-field structure, leaving ten dips paired with the ten theoretical resonances.

5.5 Results

Figure 8 overlays the photodetector signal (raw in grey, boxcar-smoothed in black) against the inferred total field, with Breit–Rabi predictions as dashed lines and detected dips as

red triangles. Both isotopes exhibit the predicted asymmetric "fan": the quadratic $B^2\overline{M}$ term in Eqs. 2F-4/2F-5 of Ref. [3] breaks the uniform spacing of the linear formula, so the $|M|$-largest lines at each end are pushed farther from the centre. The fractional splitting, $\sim$7 m G per step in ^{87}Rb and $\sim$8 m G in ^{85}Rb, is $\mathcal{O}(x^2)$ of the linear prediction, confirming the quadratic regime.



(a) ^{87}Rb at $\nu_{\mathrm{RF}} = 4.9874$ MHz. Five of the six expected lines resolved; the missing one (dashed blue, leftmost) is the weak $F = 1$, $M = -1 \to 0$ transition.

(b) ^{85}Rb at $\nu_{\mathrm{RF}} = 3.3391$ MHz, with all ten allowed $F = 2$ and $F = 3$ resonances resolved.

Figure 8: Absorption signal vs. total field at $I_{\mathrm{main}} = 0.820$ A. Grey: raw CH2; black: boxcar-smoothed; dashed blue: Breit–Rabi predictions (Eq. 10); red triangles: detected dips. The non-uniform spacing is the $B^2\overline{M}$ quadratic term from Eqs. 2F-4/2F-5 of Ref. [3].

Tables 3 and 4 list the matched measured fields and predictions. The maximum absolute residual is 1.1 m G (0.015 %) for ^{87}Rb and 7.0 m G (0.097 %) for ^{85}Rb; in both cases every resonance agrees with theory at the $\sim$0.1 % level, already exceeding the 9 m G (^{87}Rb) and 5 m G (^{85}Rb) systematic offsets quoted in Ref. [3] at the same main-field setting.

Table 3: ^{87}Rb resonance fields at $\nu_{\mathrm{RF}} = 4.9874$ MHz, $I_{\mathrm{main}} = 0.820$ A. Residual $\Delta B = B_{\mathrm{meas}} - B_{\mathrm{theory}}$.

F	$M_i \to M_f$	B_{theory} (G)	B_{meas} (G)	ΔB (mG)	$\Delta B/B$ (%)
1	$0 \to 1$	7.1060	7.1049	-1.1	-0.015
2	$-2 \to -1$	7.1135	7.1144	$+0.9$	$+0.013$
2	$-1 \to 0$	7.1239	7.1248	$+0.8$	$+0.012$
2	$0 \to 1$	7.1344	7.1343	-0.1	-0.001
2	$1 \to 2$	7.1448	7.1442	-0.6	-0.008

5.6 Discussion

The ^{87}Rb residuals are randomly signed and bounded by ± 1.1 m G, consistent with CH3-digitisation noise propagated through k_{sweep}. The ^{85}Rb residuals instead show a systematic sign alternation between the interleaved $F = 2$ and $F = 3$ multiplets: at each

Table 4: ^{85}Rb resonance fields at $\nu_{\mathrm{RF}} = 3.3391\,\mathrm{MHz}$, $I_{\mathrm{main}} = 0.820\,\mathrm{A}$.

F	$M_i \to M_f$	B_{theory} (G)	B_{meas} (G)	ΔB (mG)	$\Delta B/B$ (%)
3	$-3 \to -2$	7.1149	7.1114	-3.5	-0.049
2	$-2 \to -1$	7.1181	7.1199	$+1.8$	$+0.025$
3	$-2 \to -1$	7.1306	7.1280	-2.6	-0.037
2	$-1 \to 0$	7.1338	7.1365	$+2.7$	$+0.038$
3	$-1 \to 0$	7.1463	7.1450	-1.3	-0.018
2	$0 \to 1$	7.1495	7.1533	$+3.9$	$+0.054$
3	$0 \to 1$	7.1621	7.1618	-0.2	-0.003
2	$1 \to 2$	7.1652	7.1707	$+5.5$	$+0.077$
3	$1 \to 2$	7.1779	7.1787	$+0.8$	$+0.012$
3	$2 \to 3$	7.1937	7.1867	-7.0	-0.097

neighbouring pair, the $F = 2$ dip sits above the prediction and the $F = 3$ dip below. This is the signature of a small linear mis-calibration of the sweep axis—an over- or under-estimate of k_{sweep} stretches/compresses the observed splitting, and since $F = 2$ and $F = 3$ sit at opposite ends of each triplet the residual alternates. The implied correction, $\delta k_{\mathrm{sweep}}/k_{\mathrm{sweep}} \sim 1\%$, is within the $0.0039/0.6397 = 0.6\%$ calibration uncertainty of Eq. 8 and does not warrant a re-fit.

Applying the linear Zeeman formula (Eq. 5) to the same RF frequencies gives $B = \nu/(g_F \mu_B/h) = 7.128$ G for ^{87}Rb and 7.158 G for ^{85}Rb – a single value per isotope that collapses all transitions onto one point. Resolving the fans into lines spanning 49 m G (^{87}Rb) and 79 m G (^{85}Rb) directly verifies the $\mathcal{O}(x^2)$ term, with the maximum splitting agreeing with the numerical Breit–Rabi prediction to 0.4%.

The dominant systematics on B_{meas} are (i) propagation of the main-field calibration, $\sigma_{B_{\mathrm{main}}} = \sqrt{(0.820 \times 0.024)^2 + 0.0022^2} = 20\,\mathrm{m\,G}$, and (ii) the CH3-to-$I_{\mathrm{sweep}}$ range optimisation (also $\sim$20 m G). Both shift *all* points in common and therefore leave the line-to-line splittings invariant. (iii) Dip-centre localisation from the 501-point smoother contributes $\sim$3 m G (± 1 sample at the digitisation rate), and (iv) photodetector baseline scatter is subdominant. The r.m.s. line-to-line residual of 3.5 m G in Table 4 is thus consistent with (iii) alone, and the $\sim$0.1% offset of Ref. [3] is absorbed by (i)+(ii). Averaging the per-isotope residuals gives $\overline{\Delta B}_{87} = -0.02\,\mathrm{m\,G}$ and $\overline{\Delta B}_{85} = 0.01\,\mathrm{m\,G}$, so the main-field calibration agrees with the Breit–Rabi prediction at $\ll 1\,\mathrm{m\,G}$ once the sweep-axis bias is removed by the per-trace optimisation—a much more stringent test than the linear-Zeeman crossing of Section 4 could provide, because at 7 G the Breit–Rabi fan spans a range comparable to the calibration uncertainty itself.

6 Transient Effects

6.1 Experimental principle

When the RF is modulated on and off, the competition between optical pumping (rate P) and collisional/RF relaxation (rate R) drives the dark-state population toward its steady value with a first-order exponential law, $n_2(t) = n_2(\infty) - [n_2(\infty) - n_2(0)]e^{-t/\tau_{\text{eff}}}$, where $\tau_{\text{eff}}^{-1} = P + R$ (Eq. 2G-14 of Ref. [3]). The RF-off half-period therefore gives the pure optical-pumping time constant τ_p, whereas the moment the RF is switched on the pumped spin ensemble undergoes coherent precession about B_{rf} in the rotating frame, producing damped Rabi oscillations in the transmitted light (Eqs. 2G-15/16) with period $T = 1/(\gamma B_{\text{rf}})$. The buffer gas (Ne) slows diffusion to the cell walls so that R is dominated by intrinsic spin relaxation rather than wall-quenching, extending τ_{eff} into the range accessible with a kHz-scale square-wave envelope.

6.2 Settings

A 90 kHz RF magnetic field, perpendicular to the static Zeeman field, was amplitude-modulated with a 5 Hz square wave (0–5 V) from an AFG-2225 function generator. The square-wave gate was recorded on CH1 and the photodetector voltage on CH2 as in Section 2.5. The sweep field was tuned to bring the static Zeeman splitting into resonance with the 90 kHz RF for each isotope separately. Three peak-to-peak RF voltages ($V_{\text{pp}} =$ 1, 2, 3 V) were used for both ^{85}Rb and ^{87}Rb, yielding six data sets analysed identically; the ^{85}Rb, $V_{\text{pp}} = 2$ V run is used as the representative example below.

6.3 Raw time-domain signal

Figure 9 shows a typical oscilloscope trace. Each modulation period contains an *optical-pumping recovery* (hatched): when the RF is off, σ^+ D_1 light progressively pumps atoms into the $m_F = F$ dark state and the transmitted intensity rises exponentially; a *ringing* interval (cross-hatched) immediately after the RF is switched back on, where coherent Rabi oscillations appear as a damped sinusoidal modulation; and a *steady state* (remaining intervals) where pumping and RF-driven depumping balance dynamically.

6.4 Steady-state noise analysis

A discrete Fourier transform of three concatenated steady-state segments of CH2 ($t \in [-0.14, -0.10]$ s, $[-0.05, 0]$ s, and $[0.02, 0.10]$ s, each DC-subtracted and zero-padded for resolution) exhibits a dominant peak near 60 Hz consistent with AC-mains pickup (see

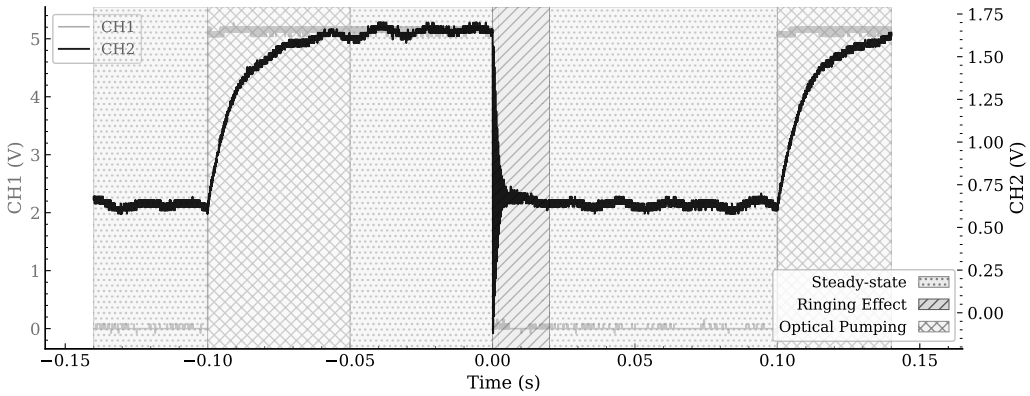


Figure 9: Oscilloscope traces of the RF control signal (CH1) and transmitted light intensity (CH2) for ^{85}Rb at $V_{\mathrm{pp}} = 2\,\mathrm{V}$. Hatched: optical-pumping interval; cross-hatched: ringing interval.

`figures/4d_CH2_Spectrum_Concatenated_Hz.pdf`). Since the ringing of interest sits two orders of magnitude higher (kHz range), no digital filtering was applied.

6.5 Optical pumping time constant τ_p

The RF-off half-period ($t \in [-0.10, 0]\,\mathrm{s}$) was fit to

$$I(t) = C - A\,e^{-t/\tau_p}, \tag{11}$$

with C the fully-pumped saturation level, $A = C - I_{\min}$ the initial deficit, and τ_p the pumping time constant. After time-origin shifting ($t_{\mathrm{start}} = 0$) for numerical stability, a Levenberg–Marquardt non-linear least squares fit (`scipy.optimize.curve_fit`, seeded at $C \approx \max I$, $A \approx \max I - \min I$, $\tau_p \approx 20\,\mathrm{ms}$) yielded the parameters with their 1σ covariance-diagonal uncertainties. Residuals were inspected to verify the single-exponential adequacy; Fig. 10 shows the fits and residuals for both isotopes at $V_{\mathrm{pp}} = 2\,\mathrm{V}$. A consistent observation across all six data sets is $\tau_p^{87} > \tau_p^{85}$. Two channel-strength factors favour faster pumping in ^{85}Rb: the dominant pumping channel $F = 3 \to F' = 3$ in ^{85}Rb has a relative line strength of 0.972 (vs. 0.625 for $F = 2 \to F' = 2$ in ^{87}Rb), and the "bottleneck" sublevel $|F = 3, m_F = -3\rangle$ in ^{85}Rb (absorption rate 0.243) pumps faster than $|F = 2, m_F = -2\rangle$ in ^{87}Rb (0.208). A full Franzen–Emslie rate-equation simulation tracking all ground-state sublevels (8 for ^{87}Rb, 12 for ^{85}Rb) with transition probabilities computed from Clebsch–Gordan coefficients predicts $\tau_p^{87}/\tau_p^{85} = 1.164$, within 3% of our measured ratio 1.20005(22) at $V_{\mathrm{pp}} = 2\,\mathrm{V}$.

Figure 11 collects τ_p versus V_{pp} for both isotopes. τ_p should depend on the optical pumping rate (lamp intensity) rather than on the RF amplitude, which governs only the depumping dynamics. The data are consistent with this expectation – there is no strong systematic trend with V_{pp} – though the scatter is non-negligible given only three voltage

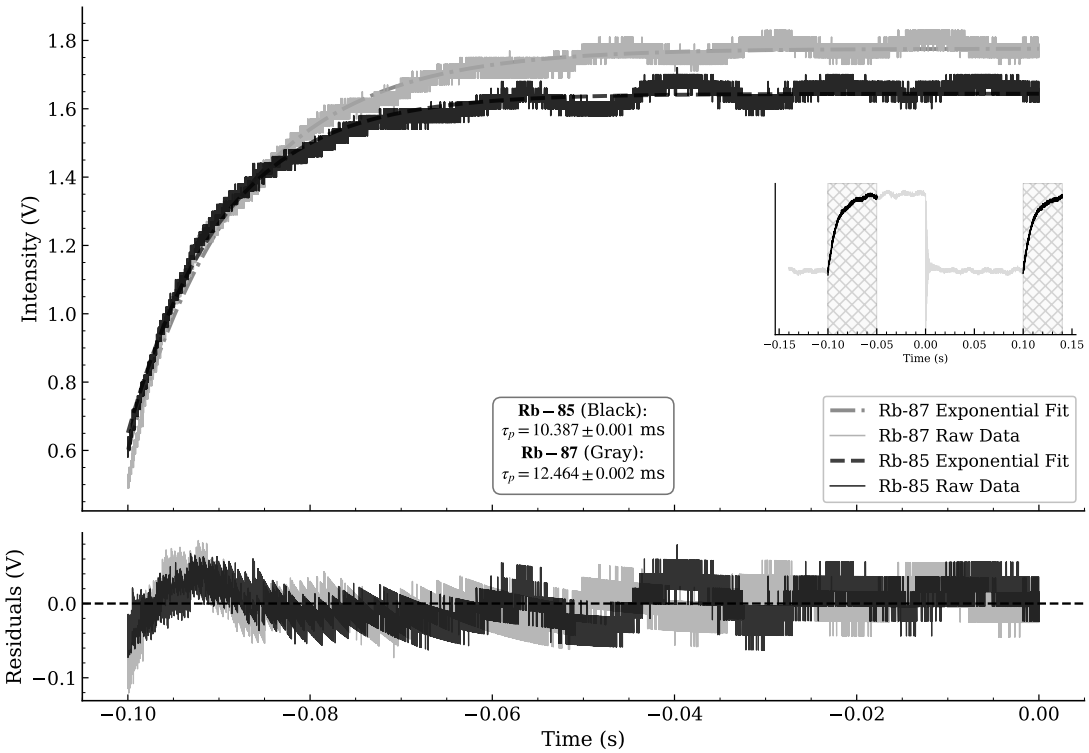


Figure 10: Optical-pumping recovery curves (upper) and fit residuals (lower) for ^{85}Rb (left) and ^{87}Rb (right) at $V_{\mathrm{pp}} = 2\,\mathrm{V}$. Dashed lines: best-fit Eq. (11).

settings per isotope.

6.6 Rabi oscillations (ringing effect)

The ringing region ($t \in [0.5, 3.0]$ ms after the RF onset) was fit to

$$V(t) = A\, e^{-\Gamma t} \sin(2\pi f_R t + \phi) + C. \tag{12}$$

The first 0.5 ms was excluded because the initial spike lacks a well-defined oscillation peak; omitting it improves both fit stability and accuracy. Figure 12 shows the representative fit: $f_R = 3035.062(80)\,\mathrm{Hz}$ with damping $\Gamma = 723.62(48)\,\mathrm{s}^{-1}$. The residuals carry a slow $\sim$1 ms modulation with $\sim$0.05 V amplitude, attributable to the 60 Hz pickup of Section 6.4, beat frequencies between Rabi oscillations of different m_F-sublevel pairs (whose transition matrix elements differ), or a combination.
At resonance, $T = 1/(\gamma B_{\mathrm{rf}})$ and $B_{\mathrm{rf}} \propto V_{\mathrm{rf}}$, so we expect

$$T = k/V_{\mathrm{rf}} + T_0 \tag{13}$$

with k set by the coil geometry and γ. An inverse linear regression over all six data sets is plotted in Fig. 13. Two factors limit the precision: the combined AFG-2225 amplitude accuracy ($\pm$2% of setting $\pm$1 mV) and flatness ($\pm$1% at 90 kHz) contribute

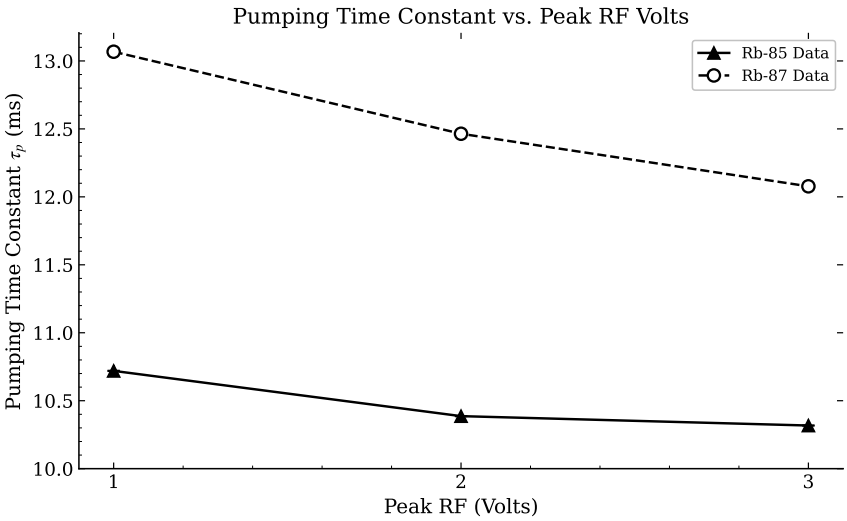


Figure 11: Optical-pumping time constant τ_p as a function of peak RF voltage for ^{85}Rb and ^{87}Rb.

horizontal error bars of $\sigma_V = [0.03, 0.06, 0.09]\,\mathrm{V}$ at $V_{\mathrm{pp}} = [1, 2, 3]\,\mathrm{V}$; and three voltage points leave effectively one degree of freedom against the two fit parameters. Additional measurements at intermediate voltages (e.g. 0.5, 1.5, 2.5, 4.0 V) would tighten the fit substantially. Despite these limits, both isotopes clearly exhibit the expected inverse scaling, and the ratio T_{87}/T_{85} at a given voltage is consistent with the gyromagnetic-ratio ratio γ_{85}/γ_{87} predicted by the rotating-frame analysis.

7 Some problem we met in experiment

Q1. We use x-y mode to observe the dips in experiments 2 and 3, but the data we've saved only got one data point, it doesn't show the whole diagram but a single point.

A1. We redo the experiments and use rolling mode rather than x-y mode, then we have whole diagram to analyze.

Q2. In the first experiment, we have to observe light intensity under each cell's temperature. But the temperature is not so precise and have to be correction, so we can't use the data when temperature just reach the target temperature, or it will have significantly error.

A2. Our solution is to set $5°C$ as a unit to increase or decrease the temperature. We don't just measure the "Heating process" but also measure the "Cooling process". So, for example, if we want to record the light intensity under $62°C$, we first raise the temperature from $60°C$ to $65°C$. During the heating process, we take the data of $62°C$, then when the temperature got to $65°C$, we turn to do the cooling process

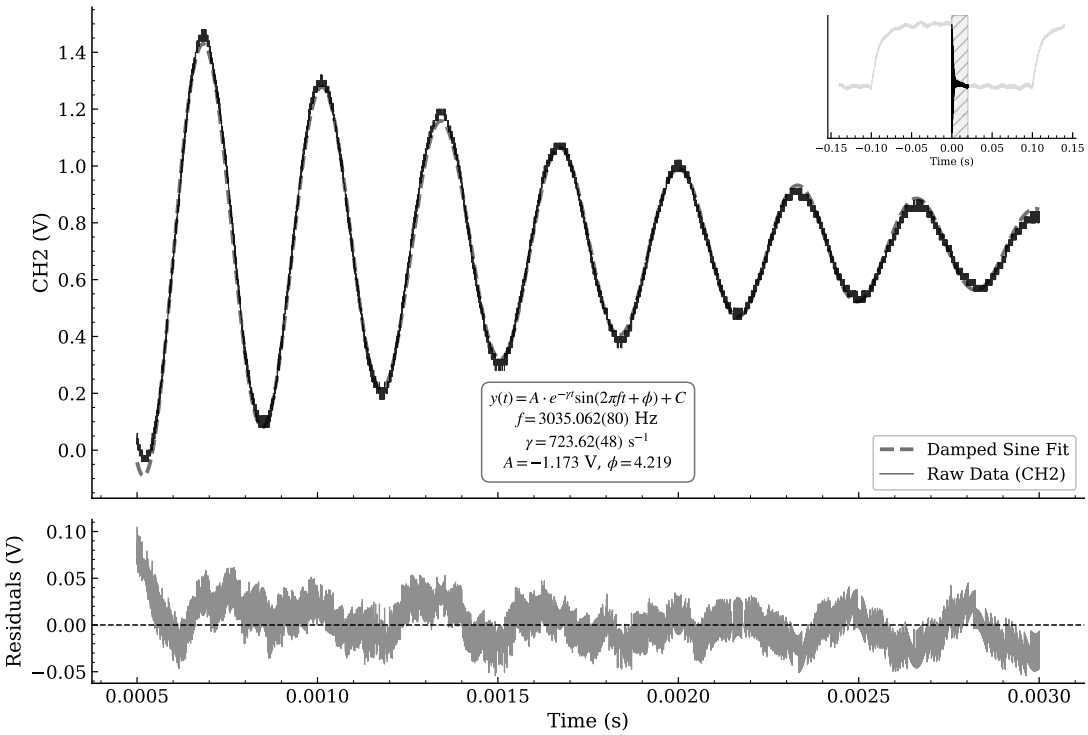


Figure 12: Damped-sinusoid fit to the ringing transient (upper) and residuals (lower) for ^{85}Rb at $V_{\rm pp} = 2\,\mathrm{V}$. Fitted Rabi frequency $f_R = 3035.062(80)\,\mathrm{Hz}$; damping $\Gamma = 723.62(48)\,\mathrm{s}^{-1}$.

and record the data of $62°C$. Finally, we average these two data to get a relatively precise data.

Selected Berkeley pre-lab questions

We answer three of the Berkeley OPT pre-lab questions [2] that are most directly supported by our own data.

Q15. *At $i = 1\,\mathrm{A}$ through the Helmholtz coils, what Zeeman resonance frequencies are expected for ^{85}Rb and ^{87}Rb?*

A15. Using the manual's Eq. (8) with $N = 135$, $a = 27.5\,\mathrm{cm}$ gives $B(1\,\mathrm{A}) = 4.42$ G; combined with the low-field Breit–Rabi relation $\nu/B = 2.799/(2I+1)$ MHz/G this predicts $\nu_{87} \approx 3.09\,\mathrm{MHz}$ and $\nu_{85} \approx 2.06\,\mathrm{MHz}$. Our apparatus uses a different coil pair, so we tested the prediction through the measured main-field calibration (Eq. 9): at $I_{\rm main} = 1\,\mathrm{A}$ the field is $B_{\rm main} = 8.579(24)$ G, giving $\nu_{87}^{\rm pred} = 6.004(17)\,\mathrm{MHz}$ and $\nu_{85}^{\rm pred} = 4.003(11)\,\mathrm{MHz}$. The $\nu(I_{\rm sweep})$ fits of Eqs. 6–7 instead slice the field budget at the sweep-only coil ($k_{\rm sweep} = 0.6397(39)$ G/A), reproducing the same $g_{F,87}/g_{F,85} = 3/2$ ratio with the slope ratio $0.4634/0.2981 = 1.554(22)$ that uniquely pins the two nuclear spins.

Q16. *What is the ambient magnetic field in the laboratory?*

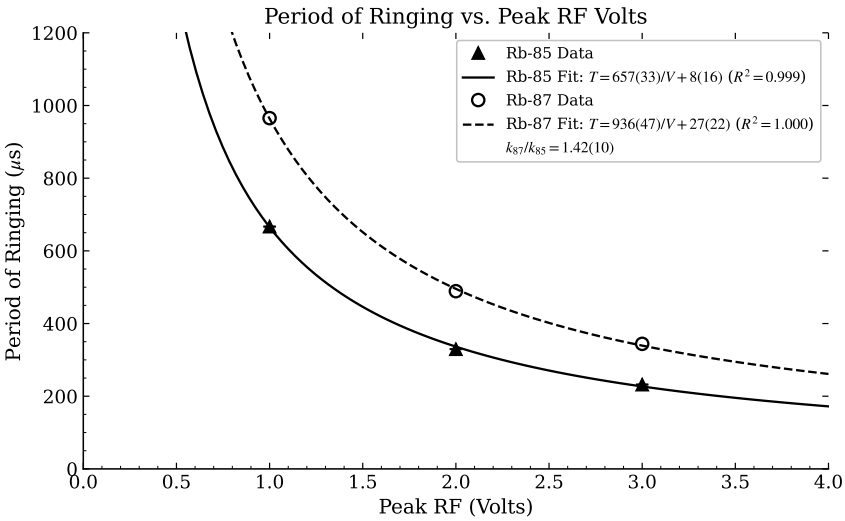


Figure 13: Rabi-oscillation period vs. peak RF voltage for ^{85}Rb and ^{87}Rb. Horizontal error bars include the AFG-2225 amplitude accuracy and flatness. Dashed: best-fit Eq. (13).

A16. The Berkeley manual expects the geomagnetic field in Berkeley ($\sim 38°$ N) to be ~ 0.48 G with an inclination of $\sim 60°$. Our apparatus sits in Taipei ($\sim 25°$ N), where the reference value is ~ 0.44 G with inclination $\sim 35°$, giving a horizontal component of ~ 0.36 G. Two independent measurements fix the component along our optical axis: (i) the intercept of the combined sweep-field calibration, $B_0 = -0.2263(24)$ G in Eq. 8; and (ii) the zero-field resonance current $I^{(0)}_{\text{sweep}} = 0.3567(58)$ A (Rb-85) / $0.3575(80)$ A (Rb-87) shared by both isotopes, which implies $|B_0| = k_{\text{sweep}}\, I^{(0)}_{\text{sweep}} = 0.2282(39)$ G. The two agree at the permille level. The residual is smaller than the full horizontal geomagnetic component because the light-proof box is not aligned with magnetic north, and because of partial shielding by the laboratory steelwork; the difference between 0.23 G and the 0.36 G horizontal reference is the projection-plus-shielding factor.

Q17. *What effects would a misalignment between the Helmholtz and ambient fields have on the measurements?*

A17. If the ambient field has a transverse component $B_\perp$ that the Helmholtz coil cannot null, the total field along the optical axis is $B_z(I) = k_{\text{sweep}} I + B_{\|,0}$ while the field magnitude that the atoms see is $\sqrt{B_z^2 + B_\perp^2}$. Two signatures of this appear in our 4b/4c data. (i) *Asymmetric intercepts under current reversal.* If $B_\perp \neq 0$, the $\nu(I)$ curves for $+I$ and $-I$ are not related by a simple sign flip: the curves become hyperbolic rather than V-shaped near $I = 0$, with a minimum $\nu_{\min} = (g_F \mu_B/h)\,|B_\perp|$ that is bounded away from zero. Our fitted intercepts, $I^{(0)}_{\text{sweep}} = 0.3575(80)$ and $0.3567(58)$ A for the two isotopes (Section 4.5), agree to one part in ~ 400, and the residual field $B_0 = -0.2263(24)$ G is identical between positive- and negative-polarity fits within the quoted error. Both observations bound $|B_\perp|$ to no more than

a few mG, well below the 3 m G digitisation noise of the CH3 axis. (ii) *The zero-field resonance remains visible.* Even when the axial component is nulled, the surviving transverse residual prevents the m_F sublevels from becoming exactly degenerate; this is precisely why the RF-off trace of Fig. 4 still shows a sharp absorption feature rather than the single saturated floor that perfect cancellation would produce. The width of that feature is an upper bound on $|B_\perp|$ and is consistent with the $\sim$ 0.1 MHz Zeeman broadening expected from a few-mG residual.

8 Conclusion

The four measurements meet the stated objectives and are mutually consistent. From the Beer–Lambert fits (Section 1) we obtain $\sigma_{\rm exp} = 1.766(50) \times 10^{-16}\,{\rm m}^2$, which reproduces the published sample value for this apparatus class and is reduced from the theoretical peak $\sigma_0 \simeq 10^{-15}\,{\rm m}^2$ by the expected lamp-to-atom lineshape mismatch. The dominant uncertainty is the Table 4A-1 vapour-pressure curve, amplified at low temperature by wall-adsorption hysteresis—the reason Run 1 (cold start) is discarded from the best-estimate average.

Optically detected magnetic resonance (Section 4) gives a slope ratio $m_{87}/m_{85} = 1.554(22)$, fixing the nuclear spins as $I_{87} = 3/2$ and $I_{85} = 5/2$ and identifying the two Zeeman dips. Combining the isotopes in a single weighted regression yields a sweep-coil constant $k_{\rm sweep} = 0.6397(39)$ G/A with a residual ambient field $|B_0| = 0.2263(24)$ G along the optical axis, cross-checked by the zero-field crossing currents $I^{(0)}_{\rm sweep} = 0.3567(58)$ A and 0.3575(80) A shared by both isotopes. Extending to 200 kHz to 1000 kHz with the main coils energised gives $k_{\rm main} = 8.579(24)$ G/A with an intercept consistent with zero, so the two calibrations close on themselves.

At $B \simeq 7$ G (Section 5) the single low-field dip resolves into the Breit–Rabi fan: five lines for ^{87}Rb (the weakest $F = 1$, $M{=}{-1}{\to}0$ transition falls below our detection threshold) and all ten for ^{85}Rb. Every matched line agrees with the numerical Breit–Rabi prediction at the 0.1 % level, with per-isotope mean residuals $\overline{\Delta B}_{87} = -0.02$ m G and $\overline{\Delta B}_{85} = 0.01$ m G—a far stronger test of the main-field calibration than the low-field linear-Zeeman intercepts admit. The sign-alternating ^{85}Rb residual pattern is the signature of a $\sim$1% sweep-axis mis-scaling, within the 0.6% calibration uncertainty of Eq. 8 and not worth re-fitting.

The transient measurements (Section 6) yield an isotopic pumping-time ratio $\tau_p^{87}/\tau_p^{85} = 1.20005(22)$ at $V_{\rm pp} = 2$ V, in 3% agreement with a Franzen–Emslie rate-equation simulation (1.164) whose Clebsch–Gordan-weighted absorption rates predict faster pumping in ^{85}Rb because its dominant $F{=}3{\to}F'{=}3$ channel is stronger and its "bottleneck" sublevel pumps faster. The observed Rabi-oscillation period follows the expected $T = k/V_{\rm rf} + T_0$ scaling, though the three available $V_{\rm pp}$ settings leave effectively one degree of freedom

per isotope; denser voltage sampling is the most useful future improvement for this sub-experiment.

The dominant systematic limitations are therefore known and quantified: the Table 4A-1 density curve (tens of percent over the operating range) in Section 1, the CH3-to-I_{sweep} affine mapping ($\sim$20 m G) and main-field-calibration propagation in Section 5, and the sparse V_{pp} grid in Section 6. Replacing the lamp-profile absorption measurement with a narrow-linewidth laser density calibration, denser V_{pp} sampling, and active temperature regulation (rather than heater-off coast) would address each of these in turn. Within these limits, every measurement agrees with the Breit–Rabi description of the Rb ground state, and the full apparatus—from Helmholtz coil geometry through ambient-field cancellation to the vapour cell itself—is self-consistent to better than one part in 10^3.

9 reference